

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Mathematics — Class IX

Syllabus: Unit 1 — Real Numbers | Unit 2 — Logarithms | Unit 4 — Factorization & Algebraic Manipulation | Unit 5 — Linear Equations & Inequalities | Unit 7 — Coordinate Geometry

ANSWER KEY (Version 1)

Total Marks: 65

Time Allowed: 2h 30m

Version: 1

SECTION A — MCQ Answer Key (12 × 1 = 12 Marks)

#	Question	Answer	Key Explanation
1	If $a = b$, property that gives $b = a$:	B	1 mark Symmetric property of equality: if $a = b$ then $b = a$. Reflexive is $a = a$ (not this); Transitive needs a third quantity c .
2	If $ac < bc$ and $c > 0$, then $a < b$ follows by:	B	1 mark Cancellation property w.r.t. multiplication : if $ac < bc$ and $c > 0$, then $a < b$. The Trichotomy property only states one of $a < b$, $a = b$, $a > b$ holds, it does not derive this from $ac < bc$.
3	Scientific notation of 0.0000728:	A	1 mark Move the decimal point 5 places right: 0.0000728 = 7.28 × 10⁻⁵ . Option C and D break the rule $1 \leq d < 10$ for the coefficient.
4	Digits in 2^{60} , given $\log 2 = 0.3010$:	C	1 mark $\log(2^{60}) = 60 \times 0.3010 = 18.06$. Characteristic = 18, so digits = $18 + 1 = 19$.
5	Antilog of 3.0000:	C	1 mark Mantissa .0000 gives table value 1.000 ; characteristic 3 places the decimal point after $3+1 = 4$ digits → antilog 3.0000 = 1000 .
6	Factors of $x^2 + 2x - 15$:	A	1 mark Need two numbers with product -15 and sum $+2$: 5 and -3 . So $x^2 + 2x - 15 = (x + 5)(x - 3)$. Check: $x^2 - 3x + 5x - 15 = x^2 + 2x - 15$ ✓
7	Factorization of $8 + y^3$:	A	1 mark Sum of cubes: $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$ with $a=2$, $b=y$ → $(2 + y)(4 - 2y + y^2)$. The middle sign of the trinomial factor is always opposite to the binomial's sign pattern (minus here), which rules out option B.
8	HCF= $(x-3)$, LCM= $(x-3)(x+2)(x-5)$; product $P \times Q$:	A	1 mark Using HCF × LCM = P × Q : product = $(x-3) \times (x-3)(x+2)(x-5) = (x-3)^2(x+2)(x-5)$.
9	Solution set of $ 6 - 3y = 0$:	A	1 mark An absolute value equals 0 only when the expression inside is 0: $6 - 3y = 0 \rightarrow y = 2$. Solution set = {2} (a single value, since $ 0 $ has no $\pm$ branch).
10	Solution of $-4x \geq 20$, $x \in \mathbb{R}$:	A	1 mark Divide both sides by -4 (negative) and reverse the inequality : $x \leq -5$.
11	Quadrant of $(-4, -(-6))$:	B	1 mark $-(-6) = 6$, so the point is $(-4, 6)$: $x < 0$, $y > 0 \rightarrow$ Second quadrant .
12	If $x < 0$, $y > 0$, quadrant of $(-x, y)$:	A	1 mark $x < 0 \Rightarrow -x > 0$. Combined with $y > 0$, both coordinates of $(-x, y)$ are positive → First quadrant .

SECTION B — Short Question Answers (11 × 3 = 33 Marks)

2(i) — Real Numbers: Bank Balance / Fabric Cost per Yard 3 marks

Opening balance = Rs. 1,75,000. Cheques written: Rs. 45,790 and Rs. 1,12,921. 1

Total withdrawn = 45,790 + 1,12,921 = **Rs. 1,58,711** 1

Current balance = 1,75,000 – 1,58,711 = **Rs. 16,289** 1

OR

Total price = Rs. 139.2 for 3.2 yards. 1

Cost per yard = 139.2 ÷ 3.2 1

= **Rs. 43.50 per yard** 1

2(ii) — Real Numbers: Balloon Cost / Banner Width & Cost 3 marks

42 balloons × Rs. 1.85 each 1

Total = 42 × 1.85 = **Rs. 77.70** 2

OR

(a) Perimeter = 2(length + width) ⇒ 10 = 2(4 + w) ⇒ 5 = 4 + w ⇒ **w = 1 yard** 2

(b) Fabric cost for the whole perimeter = 10 × Rs. 3.75 = **Rs. 37.50** 1

2(iii) — Real Numbers: Hiking Range Estimate / Reasonableness Check 3 marks

Minimum total = 7 × 5.5 = **38.5 miles** 1

Maximum total = 7 × 7.5 = **52.5 miles** 1

Since 38.5 and 52.5 both lie strictly between 35 and 55, the reasonable statement is **(b) between 35 and 55 miles.** 1

OR

Actual average = 283.4 ÷ 16.2 1

= **≈ 17.49 km/litre** 1

Since 17.49 km/l is very close to his claimed 17.5 km/l, **yes, his claim is reasonable.** 1

2(iv) — Logarithms: Find Log / Find Antilog 3 marks

(a) 5313 = 5.313 × 10³ → characteristic = 3, mantissa ≈ 0.7253 → **log 5313 ≈ 3.7253** 1.5

(b) 1109 = 1.109 × 10³ → characteristic = 3, mantissa ≈ 0.0450 → **log 1109 ≈ 3.0450** 1.5

OR

(a) 3.5636: characteristic = 3, mantissa = .5636 → table value ≈ 3.661 → shift decimal after 3+1=4 digits → **antilog 3.5636 ≈ 3661** 1.5

(b) 2.9281: characteristic = 2, mantissa = .9281 → table value ≈ 8.474 → shift decimal after 2+1=3 digits → **antilog 2.9281 ≈ 847.4** 1.5

2(v) — Logarithms/Scientific Notation Application: Kalma Pak Timing / Light Travel Time

3 marks

Total time in seconds = 5 × 10⁶ = 5,000,000 s 1

In hours = 5,000,000 ÷ 3600 ≈ **1388.9 hours** 1

In days = 1388.9 ÷ 24 ≈ 57.87 → rounding down = **57 days** (≈ 5.7 × 10¹ days) 1

OR

Time = distance ÷ speed = (9.3 × 10⁷) ÷ (1.86 × 10⁵) 1

= 5 × 10² = **500 seconds** 1

In minutes = 500 ÷ 60 ≈ **8.3 × 10⁰ minutes** (≈ 8.33 min) 1

2(vi) — Factorization: Square Root / Trinomial 3 marks

$16y^2 - 56y + 49$: first and last terms are perfect squares: $(4y)^2$ and 7^2 . 1

Middle term check: $2(4y)(7) = 56y$ ✓ so it is a perfect square trinomial = $(4y - 7)^2$ 1

Square root = $\pm(4y - 7)$ 1

OR

$2x^2 - 5x - 3$: need two numbers with product $2 \times (-3) = -6$ and sum -5 : they are -6 and 1 . 1

$2x^2 - 6x + x - 3 = 2x(x - 3) + 1(x - 3)$ 1

= $(2x + 1)(x - 3)$ 1

2(vii) — Factorization: HCF of Trinomials 3 marks

$x^2 - 2x - 8 = (x - 4)(x + 2)$ 1

$x^2 - x - 12 = (x - 4)(x + 3)$ 1

Common factor: **HCF = $(x - 4)$** 1

OR

$2x^2 + 5x - 3 = (2x - 1)(x + 3)$ 1

$x^2 + 2x - 3 = (x + 3)(x - 1)$ 1

Common factor: **HCF = $(x + 3)$** 1

2(viii) — Linear Equations: Absolute Value Equations 3 marks

Let $u = |z + 3|$. Then $u - 3 = 5 - u \rightarrow 2u = 8 \rightarrow u = 4$ 1

So $|z + 3| = 4 \rightarrow z + 3 = 4$ or $z + 3 = -4$ 1

$z = 1$ or $z = -7$. Solution set = $\{1, -7\}$ 1

OR

$3|z - 2| - 4 = -2 \rightarrow 3|z - 2| = 2 \rightarrow |z - 2| = 2/3$ 1

$z - 2 = 2/3$ or $z - 2 = -2/3$ 1

$z = 8/3$ or $z = 4/3$. Solution set = $\{8/3, 4/3\}$ 1

2(ix) — Inequalities: Compound Inequality (or) / Set-Builder Notation 3 marks

$3x + 21 < 1 - x \rightarrow 4x < -20 \rightarrow x < -5$ 1

$3x + 8 > 3 - 2x \rightarrow 5x > -5 \rightarrow x > -1$ 1

Union (joined by 'or'): solution set = $\{x \mid x \in \mathbf{R}, x < -5 \text{ or } x > -1\}$ 1

OR

(a) $N = \{1, 2, 3, \dots\}$, so $2 \leq x < 5$, $x \in N \rightarrow \{2, 3, 4\}$ 1

(b) $W = \{0, 1, 2, 3, \dots\}$, so $x \leq 4$, $x \in W \rightarrow \{0, 1, 2, 3, 4\}$ 1

(c) $-4 < x \leq 3$, $x \in \mathbf{R} \rightarrow \{x \mid x \in \mathbf{R}, -4 < x \leq 3\}$ (infinite set, cannot be listed) 1

2(x) — Coordinate Geometry: Labelling Missing Vertices 3 marks

OLMN is a square with O(0,0) and L(3,0), so side length OL = 3. 1

N lies directly above O on the y-axis: **N = (0, 3)** 1

M lies directly above L, at the same height as N: **M = (3, 3)** 1

OR

ABCD is a rectangle with sides parallel to the axes. A is at the intersection of the axes near B and D: **A = (0, 0)** 1.5

C is directly left of B (same height, $y = 2$) and directly above D (same $x = -4$): **C = (-4, 2)** 1.5

2(xi) — Coordinate Geometry: Midpoint / Hypotenuse Length 3 marks

Midpoint formula: $B = ((3+x)/2, (4+y)/2) = (-1, 7)$ 1

$(3+x)/2 = -1 \rightarrow 3+x = -2 \rightarrow x = -5$ 1

$(4+y)/2 = 7 \rightarrow 4+y = 14 \rightarrow y = 10$. So $C = (-5, 10)$ 1

OR

XY = distance from $(7,4)$ to $(7,1) = 3$ (same x , vertical segment). 1

YZ = distance from $(7,1)$ to $(-3,1) = 10$ (same y , horizontal segment); since $XY \perp YZ$, the right angle is at Y , so **XZ is the hypotenuse.** 1

$XZ = \sqrt{[(7-(-3))]^2 + (4-1)^2} = \sqrt{(10+9)} = \sqrt{109} \approx 10.44$ units 1

SECTION C — Long Question Answers ($4 \times 5 = 20$ Marks)**3(i) — Logarithms: Evaluate $\log(63/20)$ & Digits in 6^{45}** 5 marks

$63/20 = (9 \times 7)/(4 \times 5) = (3^2 \times 7)/(2^2 \times 5)$ 1

$\log(63/20) = 2\log 3 + \log 7 - 2\log 2 - \log 5$ 1

$= 2(0.4771) + 0.8451 - 2(0.3010) - 0.6990 = 0.9542 + 0.8451 - 0.6020 - 0.6990$ 0.5

$= 0.4983$ 0.5

(b) $\log 6 = \log 2 + \log 3 = 0.3010 + 0.4771 = 0.7781$ 1

$\log(6^{45}) = 45 \times 0.7781 = 35.0145 \rightarrow \text{characteristic} = 35 \rightarrow \text{digits} = 35 + 1 = 36$ 1

OR

$84/25 = (4 \times 3 \times 7)/5^2 = (2^2 \times 3 \times 7)/5^2$ 1

$\log(84/25) = 2\log 2 + \log 3 + \log 7 - 2\log 5$ 1

$= 2(0.3010) + 0.4771 + 0.8451 - 2(0.6990) = 0.6020 + 0.4771 + 0.8451 - 1.3980$ 0.5

$= 0.5262$ 0.5

(b) $\log x = 2.4771$. Mantissa .4771 exactly matches the given $\log 3 = 0.4771$, so the coefficient is 3. 1

Characteristic = 2 $\rightarrow x = 3 \times 10^2 = 300$ 1

3(ii) — Factorization Application: Veranda Dimensions / Machine Surface 5 marks

Area = $x^2 - 2x - 3 = (x - 3)(x + 1)$ 1

(a) Dimensions: length = **$(x + 1)$ m**, width = **$(x - 3)$ m** 1

(b) Perimeter = $2[(x+1)+(x-3)] = 2(2x-2) = \mathbf{(4x - 4)}$ m 1

(c) Cost of fencing = $200 \times (4x-4) = \mathbf{Rs. (800x - 800)}$ 1

(d) Cost of carpeting = $250 \times (x^2-2x-3) = \mathbf{Rs. (250x^2 - 500x - 750)}$ 1

OR

Area = $25x^2 - 30x + 9 = (5x)^2 - 2(5x)(3) + 3^2 = (5x - 3)^2$ 1

(a) Side length = **$(5x - 3)$ m** 1

(b) Boundary (perimeter) = $4(5x-3) = \mathbf{(20x - 12)}$ m 1

(c) Cost of polishing = $75 \times (25x^2-30x+9) = \mathbf{Rs. (1875x^2 - 2250x + 675)}$ 1

(d) Cost of edging 2 sides = $28 \times 2(5x-3) = \mathbf{Rs. (280x - 168)}$ 1

3(iii) — Inequalities/Equations: Double Inequality & Absolute Value 5 marks

$1 \leq 5 - 3x \leq 22$. Subtract 5 from all parts: $-4 \leq -3x \leq 17$ 1

Divide all parts by -3 (reverse inequalities): $4/3 \geq x \geq -17/3$ 1

Solution set = $\{x \mid x \in \mathbf{R}, -17/3 \leq x \leq 4/3\}$. Number line: solid dots at $-17/3$ and $4/3$, shaded segment between them. 1

(b) $|5y| = 9 \rightarrow 5y = 9$ or $5y = -9$ 1

$y = 9/5$ or $y = -9/5$ 1

OR

$5 - 3x < 11 \rightarrow -3x < 6 \rightarrow x > -2$ 1

$2x + 3 < -9 \rightarrow 2x < -12 \rightarrow x < -6$ 1

Joined by 'or': solution set = $\{x \mid x \in \mathbf{R}, x > -2 \text{ or } x < -6\}$ 1

(b) $5 - \sqrt{(2x-1)} = 0 \rightarrow \sqrt{(2x-1)} = 5 \rightarrow 2x-1 = 25 \rightarrow x = 13$ 1

Restriction: $2x-1 \geq 0 \rightarrow x \geq 1/2$ (satisfied). Check: $5 - \sqrt{25} = 5-5 = 0$ ✓ $x = 13$ is valid (not extraneous). 1

3(iv) — Coordinate Geometry: Hassan & Ali Race Problem / Rhombus Proof 5 marks

(a) Both start 6 units from the origin: Hassan is on the x-axis $\rightarrow$ **Hassan = (6, 0)**; Ali is on the y-axis $\rightarrow$ **Ali = (0, 6)**. 1

(b) C is the midpoint of Hassan(6,0) and Ali(0,6): $C = ((6+0)/2, (0+6)/2) = (3, 3)$ 2

(c) Distance Hassan $\rightarrow$ C = $\sqrt{[(6-3)^2 + (0-3)^2]} = \sqrt{(9+9)} = 3\sqrt{2}$. Distance Ali $\rightarrow$ C = $\sqrt{[(0-3)^2 + (6-3)^2]} = \sqrt{(9+9)} = 3\sqrt{2}$. Since both equal $3\sqrt{2}$, the distances are verified equal. 2

OR

Side	Calculation	Length
AB	$\sqrt{[(7-5)^2 + (5-8)^2]} = \sqrt{(4+9)}$	$\sqrt{13}$
BC	$\sqrt{[(5-7)^2 + (2-5)^2]} = \sqrt{(4+9)}$	$\sqrt{13}$
CD	$\sqrt{[(3-5)^2 + (5-2)^2]} = \sqrt{(4+9)}$	$\sqrt{13}$
DA	$\sqrt{[(5-3)^2 + (8-5)^2]} = \sqrt{(4+9)}$	$\sqrt{13}$

All four sides equal $\sqrt{13}$, so **ABCD is a rhombus**. 3

Diagonal AC = $\sqrt{[(5-5)^2 + (2-8)^2]} = \sqrt{36} = 6$. Diagonal BD = $\sqrt{[(3-7)^2 + (5-5)^2]} = \sqrt{16} = 4$. 2